

STAR Statistical Vocabularies

Automatic Extraction and Classification of Statistical Terminology

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Abstract

This project presents a pipeline for extracting and organising statistical vocabulary from a large collection of Eurostat tables. The system analyses descriptive table content and official titles to identify temporal expressions, geographical references and statistical terminology. After temporal and geographical information is separated, the remaining terms are combined into a global vocabulary and classified as measures, dimension names, dimension values, units or other terms.

The detected measures are subsequently assigned to one of seventeen statistical domains using lexical rules and TF-IDF similarity. The original dataset contained 7,605 tables. One file was excluded because its internal GZIP stream was incomplete, while the remaining 7,604 tables were processed successfully without errors.

The final vocabulary contained 162,236 distinct terms: 11,209 measures, 15,204 dimension names, 132,992 dimension values, 2,830 units and one term in the residual category. Manual evaluation achieved an accuracy of 86.42% for vocabulary classification and 85.88% for measure-domain assignment. Domain assignment also obtained a macro F1-score of 0.8459 and a weighted F1-score of 0.8717.

The results indicate that rule-based extraction and lightweight statistical methods can produce a structured statistical vocabulary from a large and heterogeneous collection of tables without requiring external generative AI services.

Keywords: statistical vocabulary, Eurostat, information extraction, semantic classification, TF-IDF, data processing.

1 Introduction

Statistical tables contain not only numerical observations, but also a large amount of descriptive information. Column names, categorical values and table titles define measures, dimensions, units, geographical areas and temporal periods. This terminology is essential for understanding the meaning of the data, but it is often distributed across many files and represented in different formats.

The objective of this project is to build a structured statistical vocabulary from a large collection of Eurostat tables. The proposed pipeline extracts temporal information, non-numeric descriptive terms and geographical entities from each table. It also processes official table titles in order to recover relevant terminology that may not appear directly in the table content.

After the extraction stage, the resulting terms are combined into a global vocabulary and classified into five categories: measures, dimension names, dimension values, units and other terms. The detected measures are then assigned to one of seventeen statistical domains. This

second classification stage combines lexical rules with TF-IDF similarity in order to provide a domain label and a confidence level for each measure.

The implementation was designed to process thousands of heterogeneous and partially compressed files while limiting memory consumption through chunk-based reading. The complete workflow is reproducible through a configuration file and a single main execution script.

The project covers Steps 1–7 of the assignment. The optional stage for extracting semantic relations between terms was not implemented. The following sections describe the data sources, the extraction methodology, the classification approach, the experimental results, the evaluation process and the main limitations of the system.

2 Data Sources and Preparation

The project uses three main input resources: the Eurostat table collection, a file that maps table names to official titles, and a NUTS workbook containing geographical nomenclature.

The complete table collection contains 7,605 CSV files. Some files use the `.csv` extension while being internally compressed with GZIP. For this reason, the input layer detects compression automatically before reading each file. Large tables are processed in blocks of 20,000 rows in order to reduce memory consumption.

The file `aact_ali01.csv` was excluded from the final execution because its internal GZIP stream was incomplete. The remaining 7,604 tables were processed successfully. Four additional files contained irregular quotation marks, but they were read by treating quotation marks as ordinary characters.

Official table titles are obtained from the file:

`file-names_to_titles_eurostat_unfiltered.csv`

This resource contains 7,678 title records. During processing, 7,604 titles were matched to the selected tables, and no processed table remained without a title.

Geographical recognition is based on the workbook:

`NUTS2021-NUTS2024.xlsx`

The workbook is transformed into a normalised geographical dictionary containing 4,512 entries. These entries include geographical codes and names used to identify regional terms in table values and titles.

All input paths are defined in `config/config.yaml`. The final execution used the complete dataset, a chunk size of 20,000 rows and relative directories for intermediate files, processed data, final outputs and logs. This configuration makes the workflow reproducible without depending on local absolute paths.

3 Vocabulary Extraction Methodology

The extraction process is applied independently to each selected table. Its objective is to separate temporal and geographical information from the remaining statistical terminology and to construct a vocabulary associated with every table.

3.1 Temporal information

The first stage identifies temporal expressions in descriptive table values. Individual years and supported year ranges are normalised as temporal intervals and stored separately from the statistical vocabulary.

For each table t , the resulting set of intervals is represented as:

$$D(t) = \{[y_{\text{start}}, y_{\text{end}}]\}$$

A single year is represented as an interval whose initial and final values are equal. The implementation is located in `src/extract_time.py`.

This stage processed all 7,604 selected tables, detected 198,090 temporal intervals and found 31 tables without temporal expressions.

3.2 Initial textual vocabulary

After temporal extraction, the system reads the descriptive columns of each table and collects their non-numeric textual values. Empty values, numerical observations and repeated terms within the same table are removed.

The resulting duplicate-free set is denoted by:

$$S(t) = \{s_1, s_2, \dots, s_n\}$$

Terms are normalised before being stored so that equivalent values can be compared across tables. The implementation in `src/extract_vocabulary.py` generated 1,151,463 distinct table-term pairs and 159,477 provisional global terms.

3.3 Geographical filtering

Geographical terms are detected by comparing the extracted values with the normalised NUTS dictionary. The detected geographical subset of each table is represented as $Geo(t)$.

These terms are stored separately and removed from the provisional statistical vocabulary:

$$V_{\text{base}}(t) = S(t) \setminus Geo(t)$$

This stage identified 413,625 geographical table-term pairs and 3,558 distinct geographical terms. After filtering, 737,838 non-geographical table-term pairs and 155,919 distinct terms remained.

The geographical processing is implemented in `src/extract_geo.py`.

3.4 Official title processing

Each processed table is matched with its official Eurostat title. Titles are normalised and analysed using the same temporal and geographical extraction criteria.

Dates and geographical expressions are removed because they are already represented separately. The remaining title content is treated as additional statistical terminology and added to the corresponding table vocabulary.

For each table, the final vocabulary is defined as:

$$V(t) = V_{\text{base}}(t) \cup T_{\text{residual}}(t)$$

where $T_{\text{residual}}(t)$ represents the relevant title terms remaining after temporal and geographical filtering.

A total of 7,604 titles were matched, with no missing title among the processed tables. Title processing extracted 1,284 temporal expressions and 335 geographical expressions. The implementation is contained in `src/process_titles.py`.

3.5 Global vocabulary construction

The final global vocabulary is obtained by combining the vocabularies of all processed tables:

$$V = \bigcup_{t=1}^{7604} V(t)$$

The process produced 745,418 final table-term pairs and 162,236 distinct vocabulary terms. These terms form the input of the semantic classification stage described in the following section.

4 Semantic Classification and Domain Assignment

4.1 Vocabulary classification

The 162,236 terms of the global vocabulary are classified into five mutually exclusive categories:

- **Measure:** statistical indicators or quantities being observed.
- **Dimension name:** labels that describe the structure of a table, such as age, sex or activity.
- **Dimension value:** categorical values associated with a dimension.
- **Unit:** expressions that describe the scale or form in which a measure is represented.
- **Other:** terms that cannot be assigned reliably to the previous categories.

The classification is implemented in `src/classify_terms.py`. It applies lexical and structural heuristics to each term and assigns exactly one category.

The rules consider patterns such as known unit expressions, typical dimension labels, categorical value formats and linguistic structures commonly associated with statistical measures. The order of the rules is relevant because some expressions may satisfy more than one condition. More specific patterns are therefore evaluated before more general ones.

The classification produces one file for each main category:

- `measures.csv`
- `dimension_names.csv`
- `dimension_values.csv`
- `units.csv`

A complete classification file and a category summary are also generated for validation and analysis.

4.2 Measure-domain assignment

After classification, the 11,209 detected measures are assigned to one of seventeen predefined statistical domains.

The assignment is implemented in `src/cluster_measures.py`. It combines domain-specific lexical rules with TF-IDF similarity. Each domain is represented by a set of descriptive keywords and expressions. The measure text is transformed into a TF-IDF vector and compared with the domain representations.

For a measure m and a domain d , the similarity score can be expressed as:

$$\text{score}(m, d) = \cos(\mathbf{v}_m, \mathbf{v}_d)$$

where $\mathbf{v}_m$ is the TF-IDF vector of the measure and $\mathbf{v}_d$ is the vector representing the domain vocabulary.

Lexical rules are used to reinforce assignments when a measure contains terms strongly associated with a specific domain. The final label corresponds to the domain with the highest combined score.

The seventeen domains are:

- agriculture, forestry and fisheries;
- business, industry and trade;
- crime, justice and safety;
- demography and population;
- economy and national accounts;
- education and training;
- energy;
- environment and climate;
- government and public finance;
- health;
- housing and construction;
- income, poverty and living conditions;
- labour market;
- science, technology and digital society;
- tourism;
- transport and mobility;
- other or multidisciplinary.

Each assignment includes a confidence level: high, medium or low. This value reflects the strength of the lexical and similarity evidence supporting the selected domain. Low-confidence predictions are retained because every detected measure must receive a domain, but they should be interpreted with greater caution.

The generated outputs include the domain assigned to each measure, the corresponding score, the confidence level, a domain summary and representative examples for each group.

5 Experimental Results

The complete execution processed 7,604 Eurostat tables. One file was excluded before execution because of an incomplete internal GZIP stream. No processing errors were reported for the selected tables.

5.1 Extraction results

Table 1 summarises the main outputs of the extraction stages.

Table 1: Main extraction results

Result	Value
Original tables	7,605
Excluded tables	1
Processed tables	7,604
Tables with errors	0
Tables without detected dates	31
Temporal intervals	198,090
Initial table-term pairs	1,151,463
Provisional global terms	159,477
Geographical table-term pairs	413,625
Distinct geographical terms	3,558
Remaining non-geographical pairs	737,838
Remaining non-geographical terms	155,919
Final table-term pairs	745,418
Final vocabulary terms	162,236

The difference between the provisional and final vocabularies is explained by the addition of relevant terms extracted from official titles. Title processing matched all 7,604 selected tables, extracted 1,284 temporal expressions and detected 335 geographical expressions.

5.2 Vocabulary categories

The final vocabulary contained 162,236 distinct terms. Table 2 shows their distribution across the five semantic categories.

Table 2: Distribution of vocabulary categories

Category	Terms	Percentage
Measures	11,209	6.91%
Dimension names	15,204	9.37%
Dimension values	132,992	81.97%
Units	2,830	1.74%
Other	1	< 0.01%
Total	162,236	100%

Dimension values form the largest group because Eurostat tables contain many categorical labels describing countries, age groups, economic activities, products and other statistical breakdowns. Measures represent 6.91% of the final vocabulary, while dimension names and units account for smaller but structurally important parts of the extracted terminology.

5.3 Domain assignment

All 11,209 measures received a statistical domain. Table 3 presents the confidence distribution.

Table 3: Confidence levels for domain assignment

Confidence	Measures	Percentage
High	5,851	52.20%
Medium	2,240	19.98%
Low	3,118	27.82%
Total	11,209	100%

More than half of the measures received a high-confidence assignment. Low-confidence results mainly correspond to short, generic or multidisciplinary expressions that do not contain enough lexical information to distinguish a specialised domain.

Table 4 shows the number of measures assigned to each domain.

Table 4: Distribution of measures by statistical domain

Domain	Measures
Other or multidisciplinary	2,844
Labour market	1,475
Science, technology and digital society	1,266
Education and training	1,042
Business, industry and trade	879
Demography and population	640
Economy and national accounts	557
Transport and mobility	462
Income, poverty and living conditions	397
Health	336
Agriculture, forestry and fisheries	279
Government and public finance	217
Environment and climate	204
Energy	180
Housing and construction	158
Tourism	138
Crime, justice and safety	135
Total	11,209

The largest group is the multidisciplinary domain, which contains measures that are too general to be assigned reliably to a more specific category. Labour market, science and technology, and education are the most frequent specialised domains in the extracted vocabulary.

6 Evaluation

The semantic stages were evaluated using manually annotated samples. Separate samples were created for vocabulary classification and measure-domain assignment in order to assess both tasks independently.

6.1 Evaluation procedure

For vocabulary classification, a sample of terms was extracted from the generated categories and manually assigned to the correct class. The predicted label was then compared with the manual

annotation.

For domain assignment, a second sample of detected measures was manually associated with the most appropriate statistical domain. The automatic domain label was compared with this reference annotation.

The evaluation uses accuracy, precision, recall and F1-score. Accuracy is defined as:

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{Total evaluated instances}}$$

For each category, precision and recall are calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Confusion matrices were also generated to analyse the most common classification errors.

6.2 Vocabulary classification evaluation

The classification sample contained 81 manually annotated terms. The system correctly classified 70 terms and incorrectly classified 11, resulting in an accuracy of 86.42%.

Table 5: Vocabulary classification evaluation

Category	Precision	Recall	F1-score	Support
Measure	1.0000	0.8000	0.8889	25
Dimension name	1.0000	1.0000	1.0000	20
Dimension value	0.8000	0.8000	0.8000	20
Unit	0.6500	1.0000	0.7879	13
Other	1.0000	0.3333	0.5000	3

Dimension names obtained perfect precision and recall in the sample. Measures also achieved high performance, although some valid measures were classified as other categories. Units obtained full recall but lower precision, which indicates that some non-unit expressions were incorrectly identified as units.

The result for the *other* category is less representative because the sample contained only three instances. Its low recall should therefore be interpreted with caution.

6.3 Domain-assignment evaluation

The domain evaluation sample contained 85 manually annotated measures. The system produced 73 correct and 12 incorrect assignments.

Table 6: Domain-assignment evaluation

Metric	Value
Evaluated measures	85
Correct assignments	73
Incorrect assignments	12
Accuracy	85.88%
Macro F1-score	0.8459
Weighted F1-score	0.8717

The difference between macro and weighted F1-score reflects the unequal number of examples across domains. More frequent domains have a greater influence on the weighted result, while the macro score gives the same importance to every domain.

Most errors occurred in generic or multidisciplinary measures with limited lexical evidence. These expressions may reasonably belong to more than one statistical area, making automatic domain assignment more difficult.

Overall, both semantic stages achieved accuracies above 85%, showing that the proposed rules and similarity-based methods provide reliable results for the tested samples.

7 Implementation and Reproducibility

The project was implemented in Python 3.11. The pipeline is divided into independent modules under the `src` directory, while `main.py` provides a single entry point for the complete execution.

The main processing stages are coordinated in `src/pipeline.py`. Temporal extraction, vocabulary extraction, geographical filtering, title processing, semantic classification and domain assignment are implemented in separate modules. This organisation simplifies testing, maintenance and the replacement of individual components.

All paths and processing parameters are defined in `config/config.yaml`. The final execution used the complete dataset, a chunk size of 20,000 rows and the exclusion of `aact_ali01.csv`. Relative paths are used throughout the project, avoiding dependencies on a specific local directory structure.

The principal final outputs are:

- `measures.csv`
- `dimension_names.csv`
- `dimension_values.csv`
- `units.csv`
- `measure_domains.csv`

Additional files contain complete classifications, summaries, examples and evaluation results. The final outputs are stored under `runs/full/outputs`, while the manually annotated evaluation samples and generated metrics are stored under `evaluation`.

The complete pipeline is executed with:

```
python main.py
```

The two evaluation stages are executed separately with:

```
python evaluate_classification.py
```

```
python evaluate_domains.py
```

The required software packages are listed in `requirements.txt`. The original Eurostat tables and external NUTS resources are not included in the repository because of their size, but the README specifies the required directory structure and configuration.

The final full execution reproduced the expected counts and completed without processing errors. This confirms that the workflow is deterministic for the supplied inputs and configuration.

8 Limitations and Discussion

The proposed system provides a practical solution for extracting and organising statistical vocabulary, but several limitations must be considered.

The first limitation is that vocabulary classification is based mainly on lexical and structural heuristics. These rules are effective for common statistical expressions, but they may fail when a term is ambiguous or when its meaning depends strongly on the table context. The same expression can act as a measure in one table and as a dimension value in another.

The distinction between units and dimension values is particularly difficult. Some expressions may describe a measurement scale, a category or both, depending on how they are used. This explains the lower precision obtained for the unit category during evaluation.

Geographical recognition is limited by the supplied NUTS resources and by the supported normalisation rules. Geographical entities outside this nomenclature, alternative names and highly abbreviated forms may remain undetected.

Domain assignment also presents limitations. Short and generic measures often contain insufficient information for a precise classification. In these cases, several domains may be plausible, and the system may assign the measure to the multidisciplinary category or produce a low-confidence result.

The evaluation samples are manually annotated but relatively small compared with the complete vocabulary. The reported metrics therefore provide useful evidence of performance, although they do not represent every possible type of statistical expression.

The optional semantic-relations stage was not implemented. As a result, the project identifies and classifies vocabulary terms but does not model relationships such as synonymy, hierarchy or conceptual association between them.

Despite these limitations, the obtained results are consistent across the complete collection. The absence of processing errors and the evaluation accuracies above 85% indicate that the proposed approach is suitable for generating a structured first version of a statistical vocabulary.

9 Conclusions

This project developed a complete pipeline for extracting, organising and evaluating statistical vocabulary from a large collection of Eurostat tables.

The workflow processed temporal expressions, non-numeric descriptive values, geographical entities and official table titles. After filtering dates and geographical terms, the remaining information was combined into a global vocabulary and classified into measures, dimension names, dimension values, units and other terms. The detected measures were then assigned to one of seventeen statistical domains.

The final execution processed 7,604 tables without errors and produced 162,236 distinct vocabulary terms. The classification stage identified 11,209 measures, 15,204 dimension names, 132,992 dimension values and 2,830 units. Domain assignment produced 5,851 high-confidence, 2,240 medium-confidence and 3,118 low-confidence predictions.

Manual evaluation showed an accuracy of 86.42% for vocabulary classification and 85.88% for measure-domain assignment. These results indicate that the combination of lexical rules, structural heuristics and TF-IDF similarity provides a reliable solution for the analysed dataset.

The main strengths of the project are its reproducibility, modular implementation and ability to process a large heterogeneous collection using limited memory. The system does not depend on external generative AI services and can be executed locally using the provided configuration and scripts.

Future work could improve context-sensitive classification, extend geographical resources, increase the size of the manually annotated samples and implement the optional extraction of semantic relationships between vocabulary terms.

10 Code and Data Availability

The source code, configuration files, evaluation samples, final outputs and documentation are available in the project repository. The original Eurostat tables and the external NUTS workbook are not included because of their size, but the required directory structure and execution instructions are described in the README.

The repository also contains the four principal vocabulary files required by the assignment: `measures.csv`, `dimension_names.csv`, `dimension_values.csv` and `units.csv`. The complete execution summary and the evaluation results are included to support reproducibility.

The source code and final project files are available at:

<https://github.com/JuanMiguelPinos/statistic-vocabularies.git>